BLOCK 2

LINEAR ALGEBRA

UNIT 5 MATRIX ALGEBRA

Structure

5.0 OBJECTIVES

After going through this unit, you should be able to

- explain the concept of determinant and various mathematical operations on determinants;
- present the properties of determinants;
- find product of two determinants;
- solve a system of simultaneous linear equations by Cramer's rule;
- explain the concept of matrix, its types, and operations on matrices;
- explain and use related concepts of submatrices and minors, orthogonal matrix, symmetric and skew-symmetric matrices, idempotent Matrix, adjoint and reciprocal matrices, trace of a matrix, transpose of matrix, non-singular matrix, inverse of a matrix;
- find inverse of a non-singular matrix;
- calculate rank of a matrix; and
- use partition matrix in carrying out efficiently matrix operations.

5.1 INTRODUCTION

Analysis of economic relationships often requires inclusion of a large number of interrelated variables into the model. Consequently, finding solutions to most of the problems, involving interrelated variables, under investigation become cumbersome. Linear algebra helps in various ways by providing means for concise representation of such problems, and solution mechanisms thereof. In the present unit, we discuss the tools of solving simultaneous equations with the aid of matrices and determinants. The following discussion starts with the concepts relating to determinants, their main features and solution techniques. The concepts and operational methods of matrices are taken up subsequently.

Though most of the textbooks on matrices and determinants discuss matrices first, and then discuss determinants. However, according to renowned mathematician L. Debnath, historically, determinants were discovered over two centuries before the discovery of matrices. Of course, the concept of matrix has been used for thousands of years in Mesopotamia, India and China for various purposes including solving systems of simultaneous equations. So, we follow the historical order in discussing these two terms: determinants first, followed by matrices.

Determinants were first used in solving systems of simultaneous equations in around 1683 almost simultaneously by Japanese mathematician Seki Kowa and German mathematician Leibnitz. The approach adopted by them, for introducing the concept of determinant, is almost the same as is explained in the very beginning of next section.

5.2 DETERMINANTS

5.2.1 Definition and Concept

If two linear equations in one variable viz. $a_1x + b_1 = 0$ and $a_2x + b_2 = 0$, with a_1, a_2, b_1, and b_2, here being assumed as non-zeroes, have the same/common solution, then $\frac{a_1}{a_2} = \frac{b_1}{b_2}$, i.e., $\frac{a_1}{b_1} = \frac{a_2}{b_2}$, i.e., $a_1b_2 - a_2b_1 = 0$. The expression $(a_1b_2 - a_2b_1)$ ***is called a determinant and is denoted by the symbol*** $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$.

Note that, a_1, a_2, b_1, b_2 are called the '***elements of the determinant***'. The elements may be zeroes also. The elements in the horizontal direction form ***'rows'*** and those in the vertical direction form ***'columns'***. The determinant $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$ has two rows and two columns. So, it is called a '***determinant of the second order***' and it has 2! = 2 terms in its expansion,

Again, if the three linear equations in two variables, viz.

$$a_1x + b_1y + c_1 = 0,$$

$$a_2x + b_2y + c_2 = 0 \text{ and}$$

$$a_3x + b_3y + c_3 = 0$$

have common solution, then, by cross multiplication, from the last two equations., we get

$$\frac{x}{b_2c_3 - b_3c_2} = \frac{y}{c_2a_3 - c_3a_2} = \frac{1}{a_2b_3 - a_3b_2}$$

$$\text{or,} \quad x = \frac{b_2c_3 - b_3c_2}{a_2b_3 - a_3b_2}, \quad y = \frac{c_2a_3 - c_3a_2}{a_2b_3 - a_3b_2}$$

$$\text{where} \quad (a_2b_3 - a_3b_2) \neq 0$$

For the solution to be common, these values of x and y must satisfy the first equation also. Hence, $a_1(b_2c_3 - b_3c_2) + b_1(c_2a_3 - c_3a_2) + c_1(a_2b_3 - a_3b_2) = 0$.

The expression $a_1b_2c_3 - a_1b_3c_2 + a_3b_1c_2 - a_2b_1c_3 + a_2b_3c_1 - a_3b_2c_1$ is denoted by the symbol

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

and has three rows and three columns.

Therefore, it is called a ‘***determinant of the third order***’ and it has 3! =6 terms; and to three out of these we prefix ‘+’ sign and to the remaining three we prefix negative sign according to the following rule.

Rule for determining signs of involved terms in the expansion of 3 x 3 determinant is given by the following figure in which the three columns of the determinant are written twice.

$$\begin{array}{cccccc} a & b & c & a & b & c \\ p & q & r & p & q & r \\ x & y & z & x & y & z \end{array}$$

$$|A| \text{ (or) } \det A = aqz + brx + cpy - ary - bpz - cqx$$

In a ‘determinant of n^{th} order’, i.e., a determinant having n rows and n columns--there are n! terms in its expansion and out of them half of the terms would be positive and rest would be negative according to the above rule, and as shown by the following expansion of a 3x 3 determinant

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

$$= a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Applications

Determinants have applications to the theory of equations, geometry, multiple integrals, differential equations and linear algebra. In geometry, determinants are closely related to areas and volumes and can be used to produce equations of lines, planes and other curves. Determinants provide new method for dealing with n-variable quadratic expressions which are generalizations of the equations for conic sections in two dimensions. All of these have, in turn, economic applications.

As an example, Jacobian (determinant) is an essential tool of mathematical analysis. It is a technique to linearize a system of non-linear equations at a point so that the properties of the linear system can be used to solve and analyze problems of non-linear system. In this context, a determinant of the form $\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \frac{\partial u}{\partial x}\frac{\partial v}{\partial y} - \frac{\partial u}{\partial y}\frac{\partial v}{\partial x}$, where u and v are two given functions of x and y, is termed as ‘Jacobian’ of u and v with respect to x and y and is often written as $\frac{\partial(u,v)}{\partial(x,y)}$.

Similarly, $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ is a 3rd order Jacobian of u, v and w, as functions of x, y and z.

5.2.2 Minors and Cofactors

It may be noted that, in the last part of previous subsection, while evaluating a 3 x 3 determinant in terms of elements of first row, a_1 is multiplied by a lower order determinant (of order 2). The second-order determinant $\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}$ is obtained by deleting the column and row containing a_1. The ***determinant*** $\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}$ ***is termed as 'minor' of the term a_1 in the original matrix***. Same procedure is applied to obtain minors of other term also. Thus, the minor of b_2 is obtained by deleting the 2^{nd} row and 2^{nd} column.

In $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ minor of $b_2 = \begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix}$

So, the minor of any element in an n^{th} order determinant is an $(n-1)^{th}$ order determinant where n=1, 2, 3, … (any finite positive number).

The 'cofactor' of any element in a determinant is its coefficient in the expansion of the determinant. It is therefore equal to the corresponding minor with a proper sign. The sign of the cofactor of an element in the i^{th} row and j^{th} column is $(-1)^{i+j}$.

Now, a_1 lies in the 1^{st} row and 1^{st} column, hence its cofactor is

$(-1)^{1+1}\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}$.

Similarly, b_3 lies in 3^{rd} row and 2^{nd} column,

Hence, its cofactor is $(-1)^{3+2}\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$.

Thus, $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1A_1 + b_1B_1 + c_1C_1 = a_1A_1 + a_2A_2 + a_3A_3$ (Do yourself)

where A_i, B_i, C_i (i=1, 2, 3) are respective cofactors of a_i, b_i and c_i

Example: Find the cofactors and the value of the following determinant

$$\begin{vmatrix} 3 & 4 & 7 \\ 2 & 1 & 3 \\ 7 & 2 & 1 \end{vmatrix}$$

Ans. Here, $a_1 = 3$, $b_1 = 4$, $c_1 = 7$, $a_2 = 2$, $b_2 = 1$, $c_2 = 3$, $a_3 = 7$, $b_3 = 2$, $c_3 = 1$

Here the cofactors are

$$A_1 = (-1)^{1+1}\begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix}$$

$$A_2 = (-1)^{1+2}\begin{vmatrix} 2 & 3 \\ 7 & 1 \end{vmatrix} = -\begin{vmatrix} 2 & 3 \\ 7 & 1 \end{vmatrix}$$

$$A_3 = (-1)^{1+3}\begin{vmatrix} 2 & 1 \\ 7 & 2 \end{vmatrix} = \begin{vmatrix} 2 & 1 \\ 7 & 2 \end{vmatrix}$$

and so on.

So, expanding the determinant by 1st row, we have its value

$$= 3\begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} - 4\begin{vmatrix} 2 & 3 \\ 7 & 1 \end{vmatrix} + 7\begin{vmatrix} 2 & 1 \\ 7 & 2 \end{vmatrix}$$

$$= 3(1-6) - 4(2-21) + 7(4-7)$$

$$= -15 + 76 - 21 = 40$$

Similarly, we can expand the determinant by any row or column. If we expand it by the 1st column, then, value of the determinant will be

$$= 3\begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} - 2\begin{vmatrix} 4 & 7 \\ 2 & 1 \end{vmatrix} + 7\begin{vmatrix} 4 & 7 \\ 1 & 3 \end{vmatrix}$$

$$= 3(1-6) - 2(4-14) + 7(12-7)$$

$$= -15 + 20 - 35 = 40$$

5.2.3 Properties of Determinants

Property I → A determinant is unchanged in value, if all the rows of the determinant are changed into columns or vice versa, i.e.,

$$\begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{21} & \dots & a_{n1} \\ a_{12} & a_{22} & \dots & a_{n2} \\ \dots & \dots & \dots & \dots \\ a_{1n} & a_{2n} & \dots & a_{nn} \end{vmatrix}$$

Example: $\begin{vmatrix} 1 & 3 & 0 \\ 2 & 4 & 0 \\ 1 & 0 & 3 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 1 \\ 3 & 4 & 0 \\ 0 & 0 & 3 \end{vmatrix}$ [following property (1)]

Property II $\rightarrow$ The value of a determinant is unaltered numerically but changed in sign if two rows (or two columns) are interchanged.

Example: $\begin{vmatrix} 1 & 3 & 0 \\ 2 & 4 & 0 \\ 1 & 0 & 3 \end{vmatrix} = -\begin{vmatrix} 3 & 1 & 0 \\ 4 & 2 & 0 \\ 0 & 1 & 3 \end{vmatrix}$ [by interchanging column (1) and (2)]

$= \begin{vmatrix} 4 & 2 & 0 \\ 3 & 1 & 0 \\ 0 & 1 & 3 \end{vmatrix}$ [by interchanging row (1) and (2)]

Property III $\rightarrow$ A determinant has zero value if the elements of one row (or of one column) are equal or proportional to the corresponding elements of a second row (or of a second column).

Example: $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{vmatrix} = 0$ [since row (1) and (3) are identical]

Example: $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 1 & 2 & 4 \end{vmatrix} = 2\begin{vmatrix} 1 & 1 & 3 \\ 2 & 2 & 1 \\ 1 & 1 & 4 \end{vmatrix} = 0$ [since column (1) and (2) are identical]

Property IV $\rightarrow$ If all the elements of any one row (or column) are multiplied (or divided) by K, then value of the determinant is also multiplied (or divided) by K, for division K is taken as non-zero.

Example:
$$\begin{vmatrix} Ka_{11} & Ka_{12} & Ka_{1n} \\ a_{21} & a_{22} & a_{2n} \\ \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & a_{nn} \end{vmatrix} = K\begin{vmatrix} a_{11} & a_{12} & a_{1n} \\ a_{21} & a_{22} & a_{2n} \\ \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & a_{nn} \end{vmatrix}$$

It can also be numerically shown by considering a 3rd order determinant

$\begin{vmatrix} 2 & 10 & 16 \\ 3 & 7 & 1 \\ 5 & 4 & 11 \end{vmatrix} = 2\begin{vmatrix} 1 & 5 & 8 \\ 3 & 7 & 1 \\ 5 & 4 & 11 \end{vmatrix}$ [since all the elements in the first row have a common factor 2]

Property V $\rightarrow$ If all the elements of any row (or column) of a determinant can be expressed as the sum of (or difference between) two numbers, then the determinant can be expressed as the sum of (or difference between) two determinants.

Example: $\begin{vmatrix} a_1+\alpha_1 & b_1+\beta_1 & c_1+\delta_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

Numerically it can be shown as follow

$$\begin{vmatrix} 8+2 & 2+5 & 5+4 \\ 4 & 9 & 1 \\ 5 & 11 & 7 \end{vmatrix} = \Delta\,(\text{say}) = \begin{vmatrix} 8 & 2 & 5 \\ 4 & 9 & 1 \\ 5 & 11 & 7 \end{vmatrix} + \begin{vmatrix} 2 & 5 & 4 \\ 4 & 9 & 1 \\ 5 & 11 & 7 \end{vmatrix} = \Delta_1 + \Delta_2 \quad (\text{say})$$

$$\therefore\ \Delta_1 = 8(63-11) - 2(28-5) + 5(44-45)$$

$$= 416 - 46 - 5 = 365$$

$$\Delta_2 = 2(63-11) - 5(28-5) + 4(44-45) = 104 - 115 - 4 = -15$$

$$\therefore\ \Delta_1 + \Delta_2 = 365 + (-15) = 365 - 15 = 350$$

and $\Delta = 10(63-11) - 7(28-5) + 9(44-45) = 520 - 161 - 9 = 350$

$\therefore\ \Delta = \Delta_1 + \Delta_2$ Hence, the proposition is proved.

Property VI $\rightarrow$ The value of a determinant remains unchanged by adding (or subtracting) k times the elements of any row (or column) to (or from) the corresponding elements of any other row (or column), where k is any given number.

Example: $\begin{vmatrix} a_1+kb_1 & b_1 & c_1 \\ a_2+kb_2 & b_2 & c_2 \\ a_3+kb_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} kb_1 & b_1 & c_1 \\ kb_2 & b_2 & c_2 \\ kb_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + k \begin{vmatrix} b_1 & b_1 & c_1 \\ b_2 & b_2 & c_2 \\ b_3 & b_3 & c_3 \end{vmatrix}$

$$= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

[since the value of the second determinant is zero as its 1st and 2nd columns are identical]

It can also be numerically shown as follows:

$$\begin{vmatrix} 2+6 & 3 & 6 \\ 3+4 & 2 & 1 \\ 4+2 & 1 & 7 \end{vmatrix} = \Delta\,(say)$$